

Title	Level of Difficulty	Score	Question Description	Answer Choice 1	Answer Choice 2	Answer Choice 3	Answer Choice 4	Correct Answer Choice	Tags
MCQ-1 Mark-1	Easy	1	When does method overloading is determined	At run time	At compile time	At coding time	At execution time	2	Polymorphism
MCQ-1 Mark-2	Easy	1	When Overloading does not occur?	different method signature	name, same signature	same signature, same name	same number of parameters	4	Polymorphism
MCQ-1 Mark-3	Easy	1	Which concept of Java is achieved by combining	Encapsulation	Inheritance	Polymorphism	Abstraction	1	Encapsulation
MCQ-1 Mark-4	Easy	1	Which keyword is used in Java to inherit a class	extends	implements	inherits	super	1	Inheritance
MCQ-1 Mark-5	Easy	1	What is an "abstract class" in Java?	A class that cannot be instantiated and must be inherited	can only contain abstract methods	without any implementation	final class that cannot be extended	1	Abstraction
MCQ-1 Mark-6	Easy	1	Rabin-Karp has a worst case complexity of $O(n^2)$	Slow hashing	log comparison at every step	causing repeated full string comparison	Large alphabet size	3	Rabin-Karp
MCQ-1 Mark-7	Easy	1	In KMP, if a mismatch occurs at pattern index j , we set j to:	0	$j - 1$	$\text{lps}[j - 1]$	$j + 1$	3	KMP
MCQ-1 Mark-8	Easy	1	In the Sieve of Eratosthenes, why do we start marking multiples from p^2 ?	p^2 is always prime	of p are already marked	To reduce array size	Multiples below p^2 are even	2	Prime Sieve
MCQ-1 Mark-9	Easy	1	What is the output of Kadane's algorithm on array $[-2, -3, -1, -4]$?	0	-10	-1	-2	3	Kadane's
MCQ-1 Mark-10	Easy	1	Using a prefix sum array, the sum of elements from index l to r can be computed in:	$O(n)$	$O(\log n)$	$O(1)$	$O(r - l)$	3	KMP
MCQ-2 Mark-11	Easy	2	What is the output of the following code snippet in Java? <pre> class Animal { public void sound() { System.out.println("Animal sound"); } } class Dog extends Animal { public void sound() { System.out.println("Bark"); } } public class Test { public static void main(String[] args) { Animal myAnimal = new Dog(); myAnimal.sound(); } } </pre>	"Animal sound"	"Bark"	Compilation error	Runtime error	2	Inheritance
MCQ-2 Mark-12	Easy	2	What is the output of this two pointer code? <pre> int[] arr = {1, 2, 3, 4, 5, 6, 7, 8, 9}; int left = 0, right = arr.length - 1; while (left < right) { int temp = arr[left]; arr[left] = arr[right]; arr[right] = temp; left++; right--; } System.out.println(arr[0] + " " + arr[4] + " " + arr[8]); </pre>	1 5 9	9 4 1	9 5 1	1 4 9	2	Window Sliding
MCQ-2 Mark-13	Easy	2	What does this print? <pre> int[] arr = {2, -1, 2, 3, 4, -1}; int currSum = 0, maxSum = Integer.MIN_VALUE; int start = 0, end = 0, tempStart = 0; for (int i = 0; i < arr.length; i++) { currSum += arr[i]; if (currSum > maxSum) { maxSum = currSum; start = tempStart; end = i; } if (currSum < 0) { currSum = 0; tempStart = i + 1; } } System.out.println(maxSum + " " + start + " " + end); </pre>	9 0 4	10 1 4	9 0 4	8 2 5	3	Window Sliding
MCQ-2 Mark-14	Easy	2	What does this Z-function snippet print? <pre> String s = "aabxaaa"; int n = s.length(); int[] z = new int[n]; int l = 0, r = 0; for (int i = 1; i < n; i++) { if (i < r) z[i] = Math.min(r - i, z[i - l]); while (i + z[i] < n && s.charAt(z[i]) == s.charAt(i + z[i])) z[i]++; if (i + z[i] > r) { l = i; r = i + z[i]; } } System.out.println(z[4] + " " + z[5]); </pre>	1 2	2 1	0 0	1 1	2	Z Function

			<pre>What does this OOP polymorphism code print? class Shape { public void draw() { System.out.println ("Drawing Shape"); } } class Circle extends Shape { public void draw() { System.out.println ("Drawing Circle"); } } class Square extends Shape { public void draw() { System.out.println ("Drawing Square"); } } public class Test { public static void main(String[] args) { Shape s = new Circle(); s.draw(); s = new Square(); s.draw(); } }</pre>					
MCQ-2 Mark-15	Easy	2	}	rawing Shape, Drawing Shaving Circle, Drawing Ciring Circle, Drawing Sc	Compile error	3	Polymorphism	

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MCQ-1 Mark-1	Easy	1	Recursion uses which memory structure internally?	Queue	Stack	Heap	Deque	2	Recursion
MCQ-1 Mark-2	Easy	1	Backtracking is mainly an extension of?	Greedy	Binary Search	Hashing	Recursion	4	Backtracking
MCQ-1 Mark-3	Easy	1	Constructor is used to?	Destroy object	Copy object	Delete object	Initialize object	4	OOPS
MCQ-1 Mark-4	Easy	1	Encapsulation refers to?	Data hiding	Searching	Sorting	Recursion	1	OOPS
MCQ-1 Mark-5	Easy	1	Insertion at head in linked list takes?	O(n log n)	O(n)	O(1)	O(log n)	3	Linked List
MCQ-1 Mark-6	Easy	1	Two pointers can be used to check?	Prime number	Binary tree height	Graph cycle	Palindrome string	4	Strings;Two Pointers
MCQ-1 Mark-7	Easy	1	Sliding window on strings is useful for finding?	Tree height	Sorting order	Graph path	Longest substring	4	Sliding Window;Strings
MCQ-1 Mark-8	Easy	1	Base case in recursion prevents?	Infinite recursion	Compilation error	Memory leak	Sorting error	1	Recursion
MCQ-1 Mark-9	Easy	1	Destructor is called when?	Function declared	Class defined	Object created	Object destroyed	4	OOPS
MCQ-1 Mark-10	Easy	1	Recursion on matrix problems commonly use?	DFS approach	Heap structure	Binary search	Sorting	1	Recursion;Matrix
MCQ-2 Mark-11	Easy	2	if (currentSum < target) { _____; } // In a sorted 2-pointer subarray sum problem, // how do we modify the window to maximize the sum?	right--	left++	left--	right++	2	Two Pointers
MCQ-2 Mark-12	Easy	2	System.out.println("programming".substring(3, 7)); // predict the output	gram	rogr	ramm	prog	1	String
MCQ-2 Mark-13	Easy	2	int l=10, r=20; int mid = l + (r - l) / 2; // Find the value of mid?	15	14	16	10	1	Binary Search
MCQ-2 Mark-14	Easy	2	class B extends A { B() { _____(); } } // Which call triggers A's constructor?	this	parent	super	new	3	OOPs
MCQ-2 Mark-15	Easy	2	sum = sum + arr[i] - arr[i - k]; // This logic belongs to which technique?	Sliding Window	Two Pointers	Binary Search	Recursion	1	Sliding Window

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MCQ-1 Mark-1	Easy	1	What is the time complexity of Kadane's Algorithm?	$O(n^2)$	$O(1)$	$O(n)$	$O(\log n)$	3	Kadane
MCQ-1 Mark-2	Easy	1	Two pointer technique reduces nested loop complexity from $O(n^2)$ to?	$O(1)$	$O(\log n)$	$O(n)$	$O(n \log n)$	3	Two Pointers
MCQ-1 Mark-3	Easy	1	Sliding Window Maximum problem is optimally solved using?	Priority Queue	Deque	Stack	Recursion	2	Deque
MCQ-1 Mark-4	Easy	1	Binary Search requires the array to be?	Positive numbers only	Sorted	Unique elements only	Unsorted	2	Binary Search
MCQ-1 Mark-5	Easy	1	Prefix sum at index i stores?	Maximum till i	Minimum till i	Sum from i to n	Sum from 0 to i	4	Prefix Sum
MCQ-1 Mark-6	Easy	1	Frequency array is commonly used instead of?	Stack	HashMap	Queue	Tree	2	Frequency Arrays
MCQ-1 Mark-7	Easy	1	Rabin Karp handles hash collisions by?	Using stack	Rechecking substring	Sorting pattern	Ignoring them	2	Rabin Karp
MCQ-1 Mark-8	Easy	1	Z-function helps compute?	Prefix matches	Subarray sum	Sorting order	Maximum element	1	Z-Function
MCQ-1 Mark-9	Easy	1	KMP algorithm runs in time complexity?	$O(\log n)$	$O(n^2)$	$O(n+m)$	$O(n \log n)$	3	KMP
MCQ-1 Mark-10	Easy	1	Recursion depth primarily depends on?	Input size	Compiler version	Operating system	IDE	1	Recursion
MCQ-2 Mark-11	Easy	2	<pre> class A { static int count = 0; } // If 5 objects of A are created, how many 'count' variables exist? Deque<Integer> stack = new ArrayDeque<>(); stack.push(1); // What is the underlying Deque operation? String s1="x"; String s2="x"; System.out.println(s1 == s2); int fib(int n) { if (n <= 1) return n; return fib(n-1) + fib(n-2); } // What is the base case is for fib(10)? while(l < r) { if(s.charAt(l) == s.charAt(r)) { l++; r--; } } // This checks for? </pre>	1	5	0	infinite	1	OOPs
MCQ-2 Mark-12	Easy	2		addLast()	addFirst()	removeFirst()	peek()	2	Deque
MCQ-2 Mark-13	Easy	2		TRUE	FALSE	Error	null	1	String
MCQ-2 Mark-14	Easy	2		$n \leq 1$	$n == 10$	$n > 1$	$n == 0$ only	1	Recursion
MCQ-2 Mark-15	Easy	2		Sorting	Palindrome	Anagram	Length	2	Two Pointers

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MCQ-1 Mark-1	Easy	1	Backtracking is commonly used in which problem?	Merge Sort	N-Queens	Binary Search	Kadane	2	Backtracking
MCQ-1 Mark-2	Easy	1	A constructor has the same name as the?	Variable	Function	Class	Object	3	OOPS
MCQ-1 Mark-3	Easy	1	Polymorphism means?	Multiple forms	Data hiding	Single behavior	Code deletion	1	OOPS
MCQ-1 Mark-4	Easy	1	Time complexity of traversing a linked list is?	O(1)	O(log n)	O(n log n)	O(n)	4	Linked List
MCQ-1 Mark-5	Easy	1	Binary Search on Answer works when there is?	Monotonic property	Unsorted data	Cycle in graph	Duplicate elements	1	Binary Search on Answer
MCQ-1 Mark-6	Easy	1	Two pointer approach on strings is useful for?	Palindrome check	Heap construction	Graph traversal	Tree balancing	1	Strings;Two Pointers
MCQ-1 Mark-7	Easy	1	Sliding window on strings is used to find?	Longest substring without repetition	Sorting order	Binary tree height	Graph shortest path	1	Sliding Window;Strings
MCQ-1 Mark-8	Easy	1	Base case in recursion ensures?	Program termination	Sorting completion	Memory allocation	Loop execution	1	Recursion
MCQ-1 Mark-9	Easy	1	Recursion on matrix problems typically use?	Binary Search	DFS	Heap	Sorting	2	Recursion;Matrix
MCQ-1 Mark-10	Easy	1	Inheritance mainly promotes?	Code reuse	Sorting efficiency	Faster execution	Memory deletion	1	OOPS
MCQ-2 Mark-11	Easy	2	<pre>class Test { private int data; } // The use of 'private' here is practical implementation</pre>	Inheritance	Polymorphism	Encapsulation	Abstraction	3	OOPs
MCQ-2 Mark-12	Easy	2	<pre>int c = (a + b) >>> 1; // This bitwise shift is equivalent to?</pre>	(a+b)/2	(a+b)*2	(a-b)/2	a+b	1	Binary Search
MCQ-2 Mark-13	Easy	2	<pre>String s1 = new String("abc"); String s2 = new String("abc"); s1.equals(s2);</pre>	FALSE	TRUE	Error	0	2	String
MCQ-2 Mark-14	Easy	2	<pre>Deque<Integer> dq = new LinkedList<>(); dq.peekFirst(); // If dq is empty, what does it return?</pre>	Exception	null	0	-1	2	Deque
MCQ-2 Mark-15	Easy	2	<pre>int sum(int n) { return n == 0 ? 0 : n + sum(n-1); } // What is the value of sum(4)?</pre>	10	6	4	15	1	Recursion

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MCQ-1 Mark-1	Easy	1	Binary Search reduces the search space by what factor each step?	Three	Half	Double	One	2	Binary Search
MCQ-2 Mark-2	Easy	1	Under which condition does Kadane's Algorithm return a maximum subarray of length 1?	Array reversed	All elements positive	Array sorted	All elements negative	4	Kadane
MCQ-3 Mark-3	Easy	1	Two pointers can detect a cycle in which data structure?	Matrix	Array	String	Linked List	4	Two Pointers
MCQ-4 Mark-4	Easy	1	Variable size sliding window is mainly used where?	Array sorted	Binary search applied	Condition based expansion	Window size fixed	3	Sliding Window
MCQ-5 Mark-5	Easy	1	Prefix sum with hashing helps solve which problem efficiently?	Two target	Binary search	Sorting	Subarray sum equals k	4	Prefix Sum/Hashing
MCQ-6 Mark-6	Easy	1	Frequency array becomes inefficient when?	Frequency array sorted	Frequency array	Range very large	Frequency array	3	Frequency Array
MCQ-7 Mark-7	Easy	1	2 array length is equal to?	n+1	n-1	n	n/2	3	2 function
MCQ-8 Mark-8	Easy	1	HashMap uses which hash recombination function?	n/2 array	PS array	PS array	Hash table	4	Hash
MCQ-9 Mark-9	Easy	1	Recursion internally uses which memory area?	Heap	Queue	Deque	Stack	4	Recursion
MCQ-10 Mark-10	Easy	1	Backtracking explores?	Greedy path	Only optimal solution	possible solutions	Random path	3	Backtracking
MCQ-11 Mark-11	Easy	2	Backtracking explores? abstract class Shape { // Shape + new Shape() // what will this look like?	Success	Compilation Error	Runtime Error	NullPointerException	2	OOps
MCQ-12 Mark-12	Easy	2	int[] arr={1, 5, 2, 8}; // What is the max sum of a sub-array of size 2? int n=arr.length-2; while(arr[n-1] swap(arr[n-1], arr[n-2]); // How does the swap function affect the array after the loop	6	7	10	13	3	Sliding Window
MCQ-13 Mark-13	Easy	2	int[] arr={1, 5, 2, 8}; // What is the max sum of a sub-array of size 2? int n=arr.length-2; while(arr[n-1] swap(arr[n-1], arr[n-2]); // How does the swap function affect the array after the loop	Sorts array	Reverses array	Clears array	Finds Max	2	Two Pointers
MCQ-14 Mark-14	Easy	2	System.out.println(20 + "Java"); int i=0, A=4, A[10]; // Binary Search index for target 8?	30Java	1020Java	Java30	Error	1	String
MCQ-15 Mark-15	Easy	2	int i=0, A=4, A[10]; // Binary Search index for target 8?	2	3	4	1	2	Binary Search

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MCD-1 Mark-1	Easy	1	Which option is used to define a generic class in Java?	[]	new	[]	<>	4	OOPS
MCD-1 Mark-2	Easy	1	Encapsulation primarily improves?	Sorting speed	Memory size	Security	Searching speed	3	OOPS
MCD-1 Mark-3	Easy	1	Linked List nodes contains?	Date and pointer	Index	Only data	Linked List	1	Linked List
MCD-1 Mark-4	Easy	1	Two pointer technique on sorted array helps find?	Median	Pair with given sum	Tree height	Graph cycle	2	Two Pointers
MCD-1 Mark-5	Easy	1	Sliding window reduces nested loops into?	Triple loop	Recursion	Single Loop	Sorting	3	Sliding Window
MCD-1 Mark-6	Easy	1	Worst case time complexity of Binary Search is?	O(n log n)	O(1)	O(n)	O(log n)	4	Binary Search
MCD-1 Mark-7	Easy	1	Maple recursion base case prevents?	Compilation error	Infinite recursion	Sorting error	Memory allocation	2	Recursion/Matlab
MCD-1 Mark-8	Easy	1	Pullman algorithm can be achieved using?	hashing	Deque	Function overloading	CORPS	4	CORPS
MCD-1 Mark-9	Easy	1	Binary Search on Answer requires?	Random array	Graph traversal	Monotonic condition	String hashing	3	Binary Search on Answer
MCD-1 Mark-10	Easy	1	Interview you problems often combine? (eg. add(int[]))	Only recursion	Only hashing	Only sorting	Multiple techniques	4	Interview Problems
MCD-2 Mark-11	Easy	2	<pre> @Override @SuppressWarnings("unchecked") public void add(E elem) { if (!isFull()) { elems.add(elem); } } </pre> System.out.println(obj.getClass().getName() + " class A : void()");	1	2	null	Error	1	Deque
MCD-2 Mark-12	Easy	2	<pre> class B extends A { void K() {} } // The method x in B is? void rec() { // Calling rec() function results in? } </pre> LIFOlike list < head, slow = head; while(list != null && list.next != null) { ... }	Overloaded	Overridden	Static	Abstract	2	OOPS
MCD-2 Mark-13	Easy	2	<pre> void rec() { // Calling rec() function results in? } </pre> LIFOlike list < head, slow = head; while(list != null && list.next != null) { ... }	Infinite Loop	StackOverflowError	Success	Memory Limit	2	Recursion
MCD-2 Mark-14	Easy	2	<pre> LIFOlike list < head, slow = head; while(list != null && list.next != null) { ... } </pre> // This is used for?	Hanging middle	Reversing	Deleting	Sorting	1	Two Pointers
MCD-2 Mark-15	Easy	2	<pre> int arrSum = right - left + 1; // If left-right=0, then endomiddle is? </pre>	3	4	5	2	2	Sliding Window